Theory behind v.0

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Derived from [1],[2],[3]&[4]

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Req. I/O according to assignment

Feature: a regression problem

-> Regression to be decided while implementing (not linear, gradient descend better)

- Input:
  - features derived from the prefix of a case
    - activities done so far
    - Timestamps
    - Attributes
- Output: a numeric value (i.e. remaining CT)

Key points:

- Attribute to select
- A uniform unit used to present / measure
- Basic idea for prediction

Critical points in basic definition

- Event logs only contain **<u>completed</u>** traces
  - Full trace
  - History
- Prefix function:

  Given a trace $\sigma=\langle e1,...,en\rangle$ and a positive integer $k\leq n$, $hd^k(\sigma)=\langle e1,...,ek\rangle$.
- Prefix log:

  Given an event log L, its prefix $logL^*$ is the event log that contains all prefixes of L, i.e.,$L^*=\{hd^k(\sigma) :\sigma\in L,1\leq k\leq|\sigma|\}$.

Critical points in basic definition (cont.)

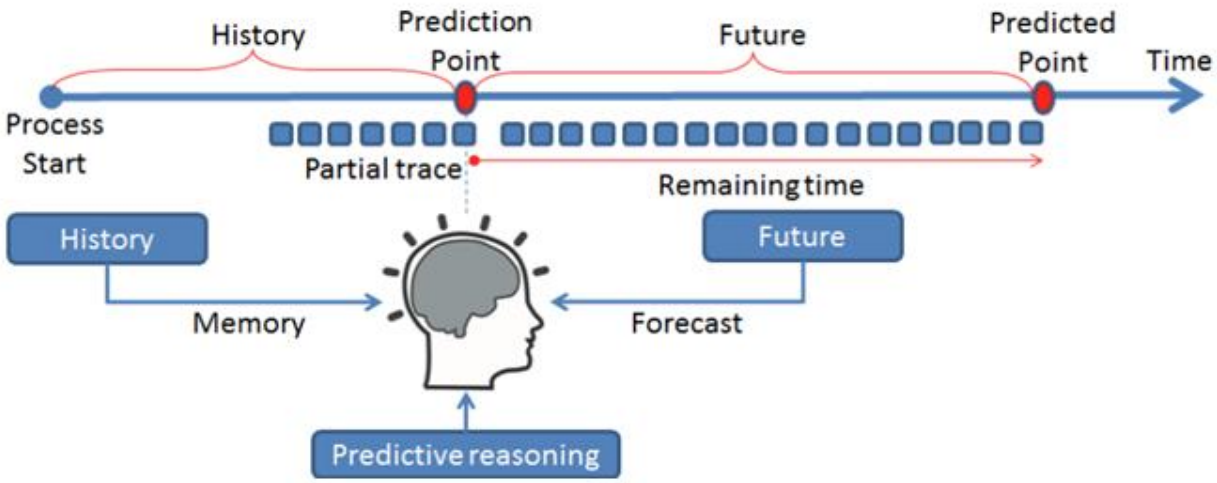


Fig. 2. Overview of predictive process monitoring.

Attribute to select (on event level)

- Mandatory (PM standard)
  - Case identifier
  - Activity name
  - Timestamp
- Two types: numeric & categorical
- More contextual attributes:
  - Resource (who’s responsible, e.g. org:role, org:group, org:resource)
    -> to detect inter-dependency & resource availability
  - lifecycle:transition (can be involved to imply queueing state)
  - Queues (e.g. how long is the queue now)

Attributes to be derived (on event level)

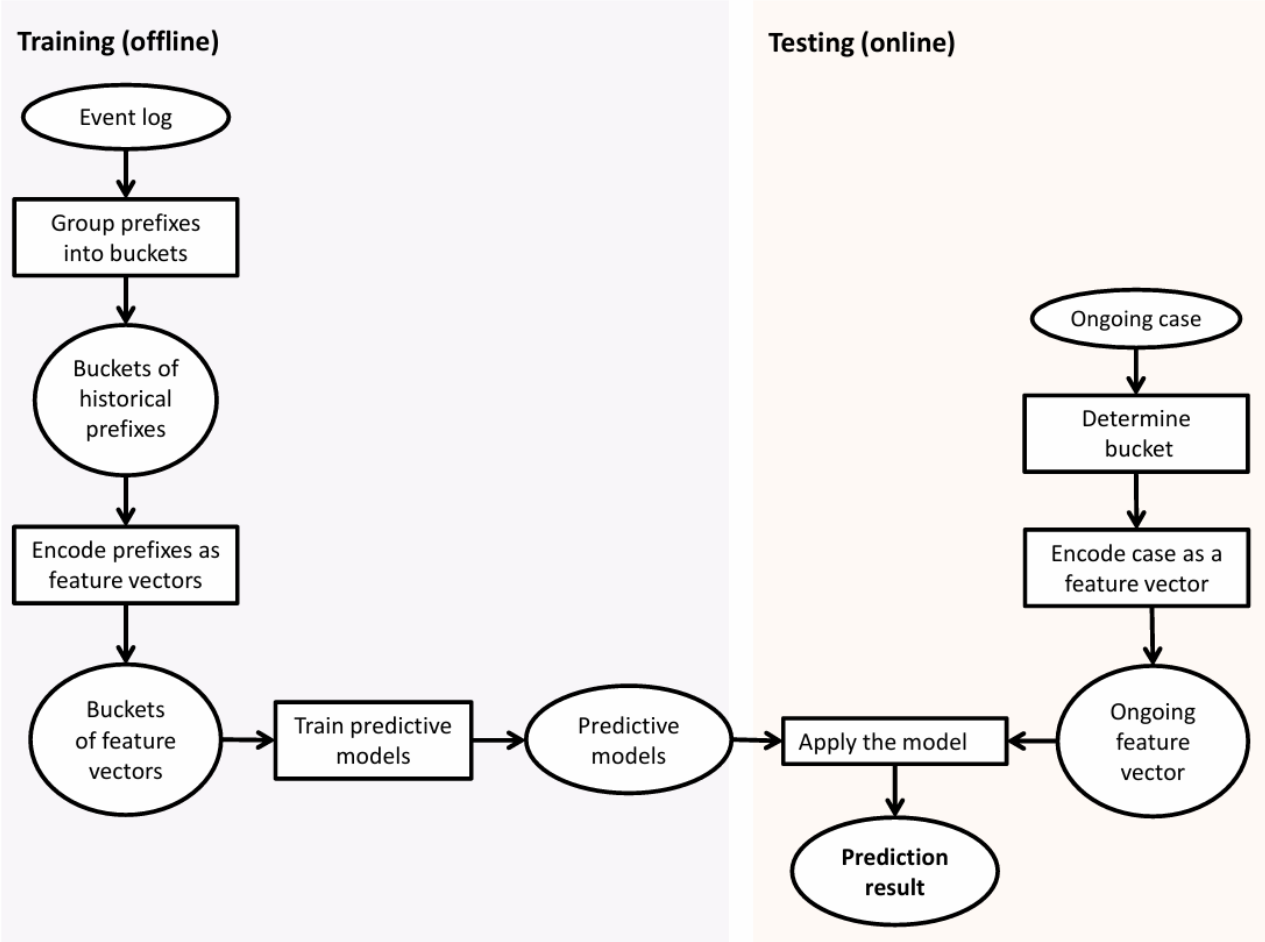
- E := Event A := Activity
- time:duration(Ex)
  - time:exiting_time(Ex) - time:entering_time(Ex)
- time:waiting_time(Ex,Ex+1)
  - time:entering_time(Ex+1) - time:exiting_time(Ex)
- int:queue_length(Ax,t)
  - |tokens with state waiting at activity Ax at time t|
- etc.

Unit used

- Hours/minutes

- Reason
  - According to the timestamp in training set:
    - Accuracy to minute-level
    - Duration between activities can be days, also can be minutes
    - To avoid overfitting & underfitting -> choose hour, but minute can also be great, further discussion needed

Basic idea for prediction



One option: XGBoost

- Generally:
  - Weighed avg. derived from several ‘weak’ decision tree learners
  - Kinda ‘forest-like’

  - For more info, cf. [2]

Options for evaluation metrics

- Accuracy:
  - Cf. [2], Mean Avg. Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

  - Root Mean Square Error
  - Mean Percentage Error

  - [2]used the first one, however, the choice should be made depending on our target event logs and defined success metrics in milestone 1

Options for evaluation metrics (cont.)

- Earliness (cf. [2])

- What we may also consider [4]:
  - For classification:
    - Brier Score loss
    - Entropy/information gain (for single decision tree)
    - Confusion matrices

Typical data quality problems to confront

- Missing data
- Incorrect data
- Imprecise data (Training data not met)
- Irrelevant data -> filter data (mainly attributes)

Other Issues

- Hard to predict rare cases

- The prediction should be ‘real-time’, i.e. it should be updated periodically/according to request from user
  - Reason: The accuracy of prediction increases as the token approaches the sink of the process

References

- [1] Wil M.P. van der Aalst. Process Mining: Data Science in Action. 2nd edition, Springer, 2016.
- [2] Ilya Verenich, Marlon Dumas, Marcello La Rosa, Fabrizio Maria Maggi, and Irina Teinemaa. “Survey and cross-benchmark comparison of remaining time prediction methods in business process monitoring”. ACM Transactions on Intelligent Systems and Technology (TIST), 10(4), 2019.
- [3]Assignment paper
- [4] https://scikit-learn.org/stable/modules/model_evaluation.html#brier-score-loss